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Department of Electronics & Communication Engineering | GATE Preparation Library

Comprehensive GATE Notes: Analog Circuits

Paper Code: EC • Section 4: Analog Circuits (8-11 Marks)

Subject Code: EC-ANA • **Target Exam:** GATE ECE • **Weightage:** 8-11 Marks

1. Operational Amplifiers: Virtual Ground & Frequency Response

- **Golden Rules of Ideal Op-Amps:** Infinite input impedance (I_+ = I_- = 0) and infinite open-loop gain (V_+ = V_- under stable negative feedback).
- **Slew Rate Constraint:** Maximum rate of change of output voltage: SR = (dV_o / dt)_max = 2*pi*f_max*V_peak. Above this frequency, sinusoidal output becomes distorted into a triangle wave.

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Inverting Op-Amp Gain	**A_v = - R_f / R_1**	R_f, R_1 = Feedback, Input Resistors	Inverting gain with virtual ground at inverting node
Non-Inverting Gain	**A_v = 1 + R_f / R_1**	Input applied directly to (+)	Non-inverting gain strictly >= 1
Slew Rate Limitation	**SR = 2 pi f_{max} V_{peak}**	SR = V/microsecond	Limits undistorted large-signal bandwidth

GATE PYQ (2 Marks) - Op-Amp Full-Power Bandwidth
Question: An inverting op-amp has R_f = 100 kOhms, R_1 = 10 kOhms, SR = 0.5 V/microsecond. For input V_in = 0.2 sin(w*t) V, find maximum frequency for undistorted output.

Step 1: |A_v| = 100k / 10k = 10.
Step 2: V_peak = 10 * 0.2 = 2.0 V.
Step 3: SR = 0.5 * 10^6 V/s = 2*pi*f*V_peak.
Step 4: f_max = (0.5 * 10^6) / (2 * pi * 2.0) = **39.79 kHz**.

2. 555 Timer Astable Multivibrator

- **Charging & Discharging Times:**
- Charging: t_high = 0.693 (R_A + R_B) C.
- Discharging: t_low = 0.693 R_B C.
- Frequency: f = 1.44 / [(R_A + 2 R_B) C]; Duty cycle = (R_A + R_B) / (R_A + 2 R_B) > 50%.

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
555 Astable Frequency	**f = 1.44 / ((R_A + 2 R_B) C)**	R_A, R_B = Timing Resistors	Oscillation frequency of free-running timer
Duty Cycle	**D = (R_A + R_B) / (R_A + 2 R_B)**	D > 0.5	Duty cycle is strictly greater than 50% without diode bypass

GATE PYQ (2 Marks) - 555 Timer Astable Frequency & Duty Cycle
Question: A 555 timer has R_A = 4 kOhms, R_B = 2 kOhms, C = 0.1 muF. Calculate oscillation frequency and duty cycle.

Step 1: f = 1.44 / [(4000 + 4000) * 10^-7] = 1.44 / (8 * 10^-4) = **1.80 kHz**.
Step 2: Duty cycle = (4 + 2) / (4 + 4) = 6/8 = **75.0%**.

Essential GATE Exam Pitfalls & Scoring Traps

- **Important:** Always check whether feedback is to (-) or (+) before assuming virtual short.
- **Important:** With positive feedback, op-amp behaves as a comparator/Schmitt trigger, not linear amplifier.